# Offline AI Assistant for Document Understanding

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# 1. Introduction



## Project Overview

An offline AI assistant intended to support document understanding on a local system.

Current project stage is foundational: the team has studied NLP and statistical language modelling and built an initial word-suggestion prototype.

## Objectives

- Build the project incrementally
- Learn the NLP and language-modelling foundations
- Develop local word-prediction capability

## Current Focus

•N-gram language models, configurable N values, probability-based suggestions, and incremental learning form the current technical foundation.

# 2. Problem Statement



## Problem

The project aims to build an AI assistant that can understand and work with documents locally, without making the final system dependent on online AI services.

## Why It Matters

- Offline operation
- Local processing of user documents
- Foundation for a privacy-conscious assistant

## Challenge Identified So Far

The team is still building the technical foundation. The weekly reports do not yet document a completed document-ingestion or document-understanding module.

# 3. Literature Review



## NLP Foundations

Week 3 focused on:

- NLP approaches
- Core NLP concepts
- Python-based NLP work

These concepts provide the base for text processing and language modelling.

## Statistical Language Models

The team learned N-gram language models and then implemented them from scratch.

N-gram models estimate the likelihood of a word using a limited preceding context.

## Gap / Limitation

Existing models involve a trade-off between simplicity, context, and computational requirements, motivating an incremental approach toward a local language model.

**Background references: [1]–[3]**

# 4. Work Plan



| Week | Work | Status |
|---|---|---|
| Week 1 | Finalized idea, objectives and scope; assigned responsibilities; initial research on technologies, tools and Python libraries; reviewed similar systems. | Completed |
| Week 2 | Set up the development environment; created GitHub repository; finalized milestones and distributed tasks. | Completed |
| Week 3 | Learned NLP approaches and fundamentals; revised Python NLP code; learned N-gram language models. | Completed |
| Week 4 | Developed a terminal-based word suggestion tool using a Bigram language model; implemented incremental learning. | Completed |
| Week 5 | Developed an N-gram language model from scratch; made N configurable for unigram, bigram, trigram and higher-order models; added probability-based suggestions. | Completed |
| Week 6 | Tested and refined the N-gram language model; verified different N values and incremental learning; analyzed model predictions and limitations. | Completed |

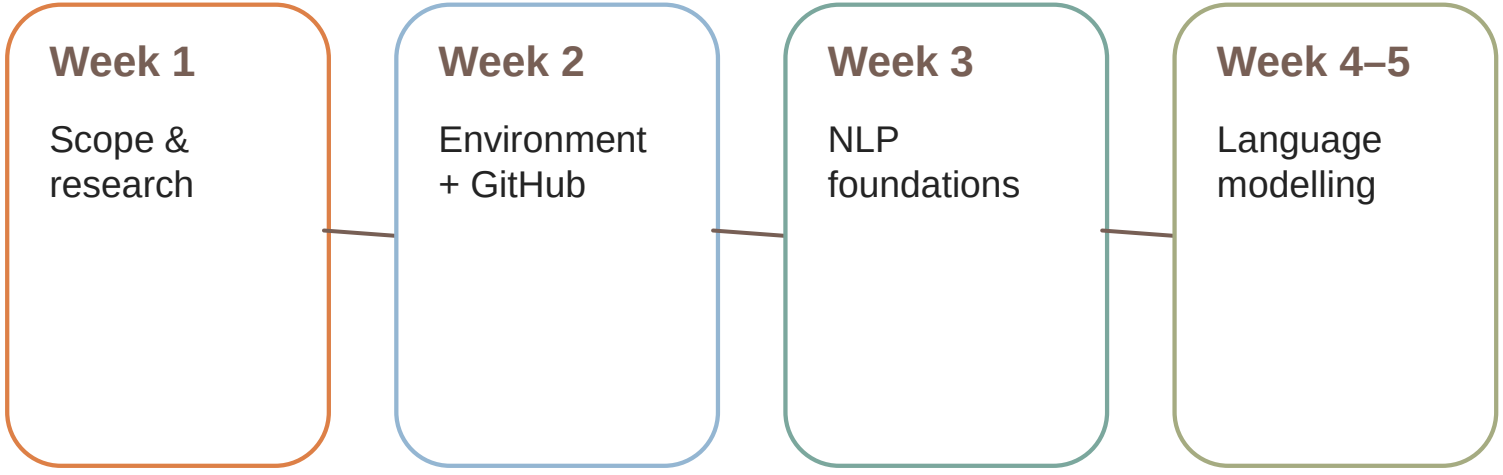
# 5. Individual Contribution



| Member Name | Enrollment No. | Documented Contribution |
|---|---|---|
| **Anurup Tiwari** | **241B607** | All three members collaboratively worked on the NLP and statistical language-modelling foundation. The team jointly implemented and tested the N-gram model, including next-word prediction and probability-based suggestions. |
| **Aarti Maheshwari** | **241B004** | All three members collaboratively worked on the N-gram model and text-processing foundation. Contributed to tokenization, text preparation and testing of the model with textual data. |
| **Khushi Sharma** | **241B327** | All three members collaboratively worked on the N-gram model and its implementation. Contributed to experimenting with different N values and testing the model. |

# 6. Work Done So Far — Foundation






**Outcome so far: the team moved from project definition and environment setup to hands-on NLP and statistical language-model implementation.**

# 6. Work Done So Far — Current Technical Design



## Structural View (proposed)

**Document**
file / text source

**TextProcessor**
tokenize & prepare text

**NGramModel**
store counts + probabilities

**SuggestionEngine**
rank next-word candidates

## Behavioral View (proposed)

1. **Load text**
2. **Tokenize**
3. **Build / update N-gram model**
4. **Calculate probabilities**
5. **Return ranked suggestions**

**Current evidence: the N-gram model and word-suggestion components have been implemented; document ingestion and the final assistant layer remain pending.**

# 6. Work Done So Far — Word Suggestion Prototype



## Terminal-Based Prototype

Week 4:

- Developed a terminal-based word suggestion tool
- Used a Bigram language model
- Added incremental learning

This is the first concrete working prototype reported by the team.

**Terminal workflow (illustrative)**

```
> Enter text
> Build / update bigram counts
> Read previous context
> Predict next word

Suggestions:
1. candidate A
2. candidate B
3. candidate C
```

# 6. Work Done So Far — N-gram Model



## From Scratch

Week 5:

Implemented an N-gram language model from scratch to understand statistical language modelling and word prediction.

## Configurable N

N = 1 → Unigram
N = 2 → Bigram
N = 3 → Trigram
N > 3 → Higher-order N-gram

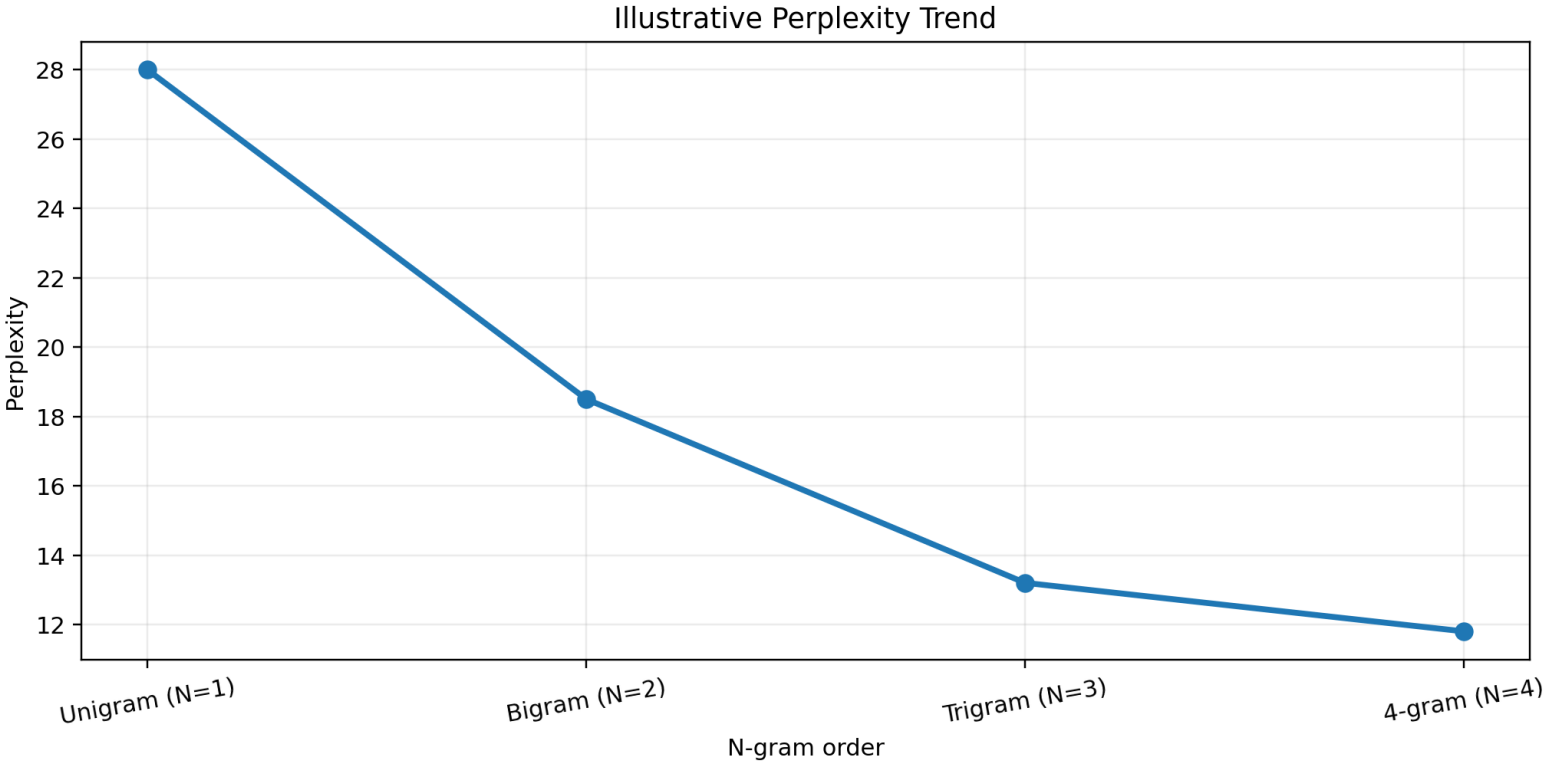
## Probability-Based Suggestions

The model was extended to suggest words using estimated probabilities rather than only returning raw frequency counts.

This provides the statistical base for later assistant capabilities.

# Model Evaluation — Perplexity






## What is Perplexity?

### Meaning

Perplexity measures how well a language model predicts unseen text.

### Interpretation

Lower perplexity generally means better predictive performance on the same test set.

### Formula

PP = exp(− average log probability)

### Project status

Perplexity evaluation is planned as part of model evaluation. The graph uses illustrative sample values, not measured project results.

## Perplexity Formula

$$PP = \exp( -(1/N) \Sigma \log P(w_i \mid context) )$$

Lower perplexity generally indicates better prediction on unseen/test text.

# 7. Work to be Done



| | | |
|---|---|---|
| 1 | **Strengthen the language model** | Continue improving the N-gram implementation and evaluate word suggestions. |
| 2 | **Move toward document input** | Introduce document ingestion and text extraction as the next application layer. |
| 3 | **Document understanding** | Build the actual local document-understanding functionality. |
| 4 | **Offline assistant layer** | Explore the supervisor-suggested direction: local LLM + RAG + domain-specific LLM. |
| 5 | **UI + evaluation** | Develop a user interface and evaluate the system with suitable test documents. |

# 8. Conclusion



## Progress So Far

- Project scope and milestones finalized
- Development environment and GitHub repository set up
- NLP fundamentals studied
- Bigram word-suggestion prototype developed
- Incremental learning implemented
- Configurable N-gram model developed
- Probability-based suggestions added

**Probability Calculation**

Bigram: $P(w_i \mid w_{i-1}) = \text{Count}(w_{i-1}, w_i) / \text{Count}(w_{i-1})$

General N-gram: $P(w_i \mid \text{context}) = \text{Count}(\text{context}, w_i) / \text{Count}(\text{context})$

The model uses these conditional probabilities to rank possible next words.

## Current Position

The project has established its NLP and statistical language-modelling foundation, but the final document-understanding assistant has not yet been built.

## Expected Outcome

A local/offline assistant capable of processing documents and supporting document understanding, built on the foundation developed during Part-I.

# 9. References



[1] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed., draft, 2025.
[2] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval. Cambridge, U.K.: Cambridge Univ. Press, 2008.
[3] S. Bird, E. Klein, and E. Loper, Natural Language Processing with Python. Sebastopol, CA, USA: O'Reilly Media, 2009.
[4] Python Software Foundation, "Python Documentation," Python 3 documentation.

References [1]–[3] provide the NLP and language-modelling background; [4] supports the Python implementation environment used during development.

# Thank You

## Offline AI Assistant for Document Understanding

Minor Project Part-I • Jaypee University of Engineering & Technology, Guna